

The diagram illustrates the sulfur cycle, showing the interconversions of various sulfur species and the enzymes involved. The cycle is organized around a central point with several sulfur species and their corresponding redox states:

- Top:** HS^- (Hydrogen sulfide)
- Top Right:** $\text{S}(0)$ (Elemental sulfur)
- Right:** SO_3^{2-} (Sulfite)
- Bottom Right:** SO_4^{2-} (Sulfate)
- Bottom:** SO_2 (Sulfur dioxide)
- Bottom Left:** SO_3^{2-} (Sulfite)
- Left:** $\text{S}_2\text{O}_3^{2-}$ (Thiosulfate)
- Top Left:** SO_3^{2-} (Sulfite)

The cycle is driven by several key enzymes and processes:

- Top Left:** DsrMKJOP (Dissimilatory sulfide reductase complex) converts HS^- to $\text{S}_2\text{O}_3^{2-}$ and SO_3^{2-} to $\text{S}_2\text{O}_3^{2-}$ (indicated by e^-).
- Top Right:** FccAB/Sqr (Ferredoxin-dependent sulfide:quinone reductase) converts HS^- to $\text{S}(0)$.
- Right:** DsrMKJOP (Dissimilatory sulfide reductase complex) converts $\text{S}(0)$ to SO_3^{2-} (indicated by e^-).
- Bottom Right:** sHdrAB1B2B3C1C2 , DsrABCEFH , and Sdo (Dissimilatory sulfite reductase complex) convert SO_3^{2-} to SO_4^{2-} .
- Bottom:** SoeABC , SorAB , and SorT (Sulfate reductase complex) convert SO_4^{2-} to SO_2 .
- Bottom Left:** Sat (Sulfate adenylyltransferase) converts SO_4^{2-} to SO_3^{2-} .
- Left:** QmoABC (Quinolinate monooxygenase) converts SO_3^{2-} to $\text{S}_2\text{O}_3^{2-}$ (indicated by e^-).
- Top Left:** AprAB (Adenosine phosphoribosyltransferase) converts $\text{S}_2\text{O}_3^{2-}$ to SO_3^{2-} (indicated by e^-).
- Central Interconversions:**
 - $\text{S}_2\text{O}_3^{2-}$ and SO_3^{2-} (Left) are interconverted by PhsABC (Phosphotransferase system).
 - $\text{S}_2\text{O}_3^{2-}$ and SO_2 are interconverted by SoxYZXAB (Sulfur oxidation system).
 - SO_2 and SO_4^{2-} are interconverted by TsdA/DoxDA (Thiosulfate desulfurase) and Ttr/Otr (Thiosulfate reductase).
 - SO_2 and SO_3^{2-} (Right) are interconverted by TetH (Thiosulfate reductase) and SoxB (Sulfur oxidation system).
 - SO_3^{2-} (Right) and SO_4^{2-} are interconverted by SoxYZXABCD (Sulfur oxidation system).

Dissimilatory sulfate reduction pathway

The diagram illustrates the dissimilatory sulfate reduction pathway, showing the flow of electrons and the location of various components relative to the cell membrane.

Periplasm:

- SO_4^{2-} is reduced to **APS** by the enzyme **Sat**.
- APS** is reduced to SO_3^{2-} by the enzyme **AprBA**.

Membrane:

- QmoABC** and **DsrMKJOP** are embedded in the membrane. Both facilitate the reduction of MQH_2 to MQ .
- QmoABC** is coupled with **AprBA** via a dashed arrow representing electron flow (e^-).
- DsrMKJOP** is coupled with **DsrC** via a solid arrow representing electron flow.

Cytoplasm:

- SO_3^{2-} is reduced to HS^- by the enzyme **DsrBA**.
- HS^- is oxidized to $\text{S}^{(0)}$ by the enzyme **DsrC**.
- $\text{S}^{(0)}$ is further oxidized to $\text{S}_2\text{O}_3^{2-}$ by the enzyme **DsrC**.

Legend:

- Periplasm:** Top region (light blue background).
- Membrane:** Middle region (green background).
- Cytoplasm:** Bottom region (light blue background).

Diagram illustrating the dissimilatory sulfide oxidation pathway, showing the flow of sulfur species across the cell envelope (Periplasm, Membrane, Cytoplasm).

Periplasm:

- FccAB:** A cycle involving HS^- and S^0 . HS^- is oxidized to S^0 (indicated by a red arrow), and S^0 is reduced back to HS^- (indicated by a black arrow).
- Sqr:** A cycle involving HS^- and S_n^{2-} . HS^- is oxidized to S_n^{2-} (indicated by a green arrow), and S_n^{2-} is reduced back to HS^- (indicated by a black arrow).

Membrane:

- DsrMKJOP:** A cycle involving MQ and MQH_2 . MQ is reduced to MQH_2 (indicated by a green arrow).
- QmoABC:** A cycle involving MQ and MQH_2 . MQ is reduced to MQH_2 (indicated by a green arrow).

Cytoplasm:

- DsrC:** A cycle involving S^0 and S^{2-} . S^0 is reduced to S^{2-} (indicated by a black arrow), and S^{2-} is oxidized back to S^0 (indicated by a black arrow).
- DsrBA and DsrL:** These proteins are involved in the reduction of S^0 to S^{2-} .
- AprBA:** A protein involved in the reduction of S^0 to S^{2-} .
- Sat:** A protein involved in the reduction of S^0 to S^{2-} .

Overall Pathway:

The pathway shows the conversion of HS^- to S^0 in the periplasm, followed by the reduction of S^0 to S^{2-} in the cytoplasm. The final product is SO_4^{2-} .

Legend:

- Red:** Oxidation
- Green:** Reduction
- Black:** Unknown

Source: Weissgerber et al. (2014), Gwak et al. (2022)